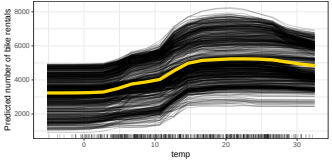
# Interpretable Machine Learning

# Partial Dependence (PD) plot



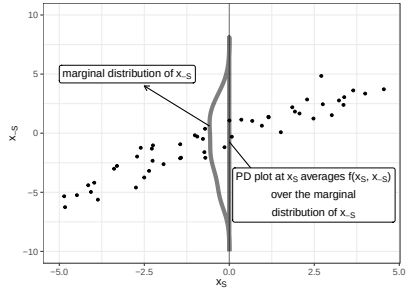

**Learning goals**

- PD plots and relation to ICE plots
- Interpretation of PDP
- Extrapolation and Interactions in PDPs
- Centered ICE and PDP

# PARTIAL DEPENDENCE (PD) ▸ Friedman (2001)

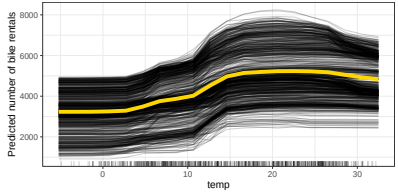
**Definition:** PD function is expectation of $\hat{f}(\mathbf{x}_S, \mathbf{x}_{-S})$ w.r.t. marginal distribution of features $\mathbf{x}_{-S}$:

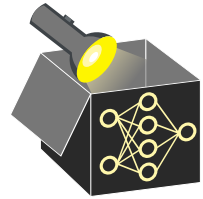
$$f_{S,PD}(\mathbf{x}_S) = \mathbb{E}_{\mathbf{x}_{-S}}\left(\hat{f}(\mathbf{x}_S, \mathbf{x}_{-S})\right)$$
$$= \int_{-\infty}^{\infty} \hat{f}(\mathbf{x}_S, \mathbf{x}_{-S})\, d\mathbb{P}(\mathbf{x}_{-S})$$




**Estimation:** For a grid value $\mathbf{x}_S^*$, average ICE curves point-wise at $\mathbf{x}_S^*$ over all observed $\mathbf{x}_{-S}^{(i)}$:

$$\hat{f}_{S,PD}(\mathbf{x}_S^*) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}(\mathbf{x}_S^*, \mathbf{x}_{-S}^{(i)})$$
$$= \frac{1}{n}\sum_{i=1}^{n} \hat{f}_{S,ICE}^{(i)}(\mathbf{x}_S^*)$$



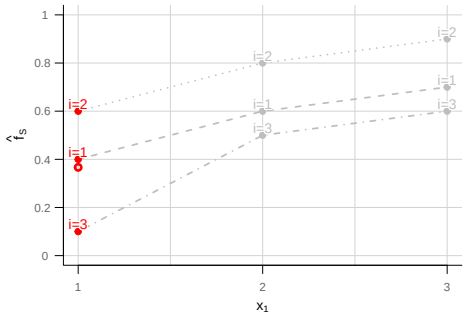
# PARTIAL DEPENDENCE

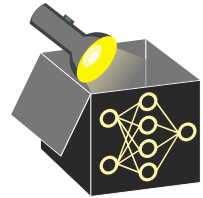
| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ | $\hat{f}$ |
|---|---|---|---|---|
| 1 | 1 | 4 | 7 | 0.4 |
| 2 | 1 | 5 | 8 | 0.6 |
| 3 | 1 | 6 | 9 | 0.1 |

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ | $\hat{f}$ |
|---|---|---|---|---|
| 1 | 2 | 4 | 7 | 0.6 |
| 2 | 2 | 5 | 8 | 0.8 |
| 3 | 2 | 6 | 9 | 0.5 |

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ | $\hat{f}$ |
|---|---|---|---|---|
| 1 | 3 | 4 | 7 | 0.7 |
| 2 | 3 | 5 | 8 | 0.9 |
| 3 | 3 | 6 | 9 | 0.6 |

| $\frac{1}{3}\sum_{i=1}^{3}\hat{f}$ |
|---|
| ⅓ (0.4 + 0.6 + 0.1) |
| ⅓ (0.6 + 0.8 + 0.5) |
| ⅓ (0.7 + 0.9 + 0.6) |

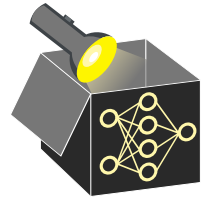


Estimate PD function by **point-wise** average of ICE curves at grid value $\mathbf{x}_S^* = x_1^* = 1$:

$$\hat{f}_{1,PD}(x_1^*) = \frac{1}{n}\sum_{i=1}^{n}\hat{f}(x_1^*, \mathbf{x}_{2,3}^{(i)})$$

# PARTIAL DEPENDENCE



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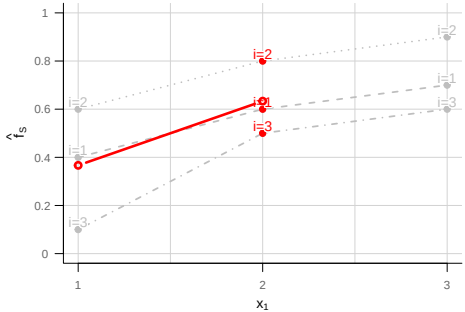


Estimate PD function by **point-wise** average of ICE curves at grid value $\boxed{\mathbf{x}_S^* = x_1^* = 2}$:

$$\hat{f}_{1,PD}(x_1^*) = \frac{1}{n}\sum_{i=1}^{n}\hat{f}(x_1^*, \mathbf{x}_{2,3}^{(i)})$$

# PARTIAL DEPENDENCE

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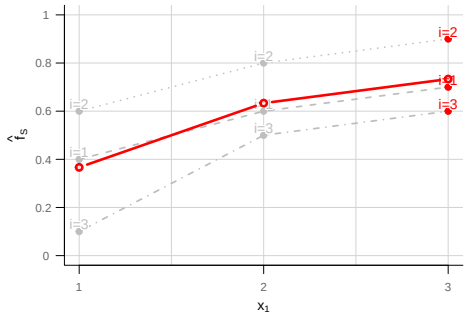
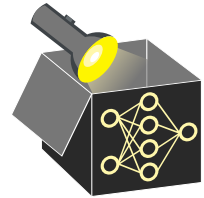
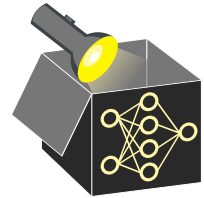


Estimate PD function by **point-wise** average of ICE curves at grid value $\boxed{\mathbf{x}_S^* = x_1^* = 3}$:

$$\hat{f}_{1,PD}(x_1^*) = \frac{1}{n}\sum_{i=1}^{n}\hat{f}(x_1^*, \mathbf{x}_{2,3}^{(i)})$$

# EXAMPLE: PD FOR LINEAR MODEL



Assume a linear regression model with two features:

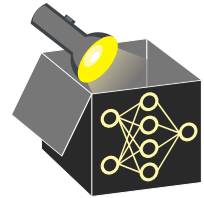
$$\hat{f}(\mathbf{x}) = \hat{f}(\mathbf{x}_1, \mathbf{x}_2) = \hat{\theta}_1 \mathbf{x}_1 + \hat{\theta}_2 \mathbf{x}_2 + \hat{\theta}_0$$

PD function for feature of interest $S = \{1\}$ (with $-S = \{2\}$) is:

$$\begin{aligned} f_{1,PD}(\mathbf{x}_1) = \mathbb{E}_{\mathbf{x}_2}\left(\hat{f}(\mathbf{x}_1, \mathbf{x}_2)\right) &= \int_{-\infty}^{\infty} \left(\hat{\theta}_1 \mathbf{x}_1 + \hat{\theta}_2 \mathbf{x}_2 + \hat{\theta}_0\right) \, d\mathbb{P}(\mathbf{x}_2) \\ &= \hat{\theta}_1 \mathbf{x}_1 + \hat{\theta}_2 \cdot \int_{-\infty}^{\infty} \mathbf{x}_2 \, d\mathbb{P}(\mathbf{x}_2) + \hat{\theta}_0 \\ &= \hat{\theta}_1 \mathbf{x}_1 + \underbrace{\hat{\theta}_2 \cdot \mathbb{E}_{\mathbf{x}_2}(\mathbf{x}_2) + \hat{\theta}_0}_{:=const} \end{aligned}$$

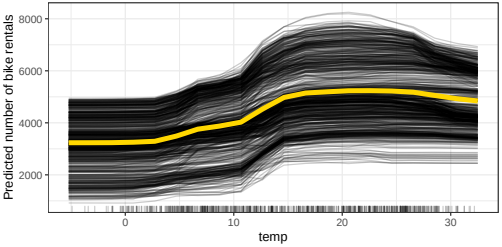
$\Rightarrow$ PD plot visualizes the function $f_{1,PD}(\mathbf{x}_1) = \hat{\theta}_1 \mathbf{x}_1 + const$ ($\hat{=}$ feature effect of $\mathbf{x}_1$).

# INTERPRETATION: PD AND ICE

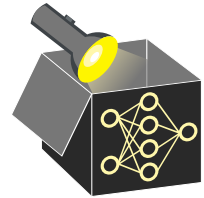


If feature varies:

- **ICE:** How does **prediction of individual observation** change? $\Rightarrow$ **local** interpretation
- **PD:** How does **average effect** / **expected prediction** change? $\Rightarrow$ **global** interpretation



# INTERPRETATION: PD AND ICE



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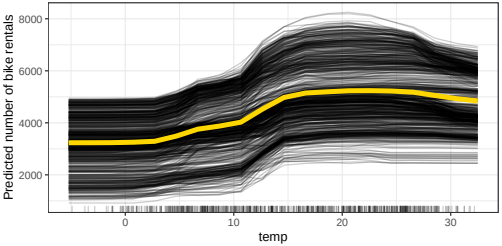
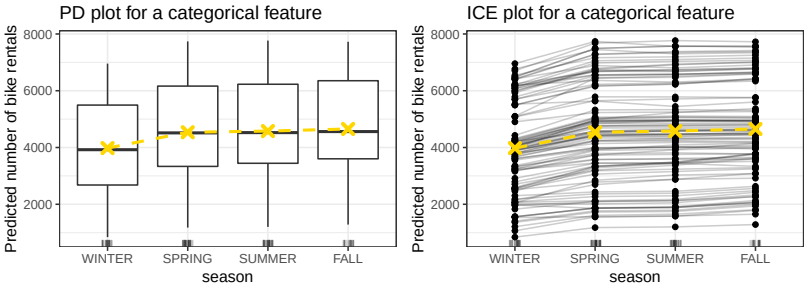
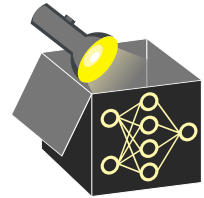



Insights from bike sharing data:

- Parallel ICE curves = homogeneous effect
- Warmer ⇒ more rented bikes
- Too hot ⇒ slightly less bikes

# INTERPRETATION: CATEGORICAL FEATURES



- PDP with boxplots and ICE with parallel coordinates plots
- NB: Categories can be unordered, if so, rather compare pairwise